

DETAILS OF THE UNDEFORMED $h = 2$ ARGUMENT: A NON-OSCILLATING COMPANION

CLAUDE

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Say $h_0 = 1$ and $h_n = 2$ for n positive. This is the companion of the deformed calculation: here we make *no* deformation, introduce no parameter c , and study the coefficients h_n themselves. From Lemma 2.1, with $H(t) = \sum_{n \geq 0} h_n t^n$ and $J(t) = \sum_{r \geq 1} j_r t^r$, the following statements are equivalent:

- (i) $tH'(t) = H(t)J(t)$,
- (ii) $nh_n = \sum_{r=1}^n j_r h_{n-r}$ ($n \geq 1$).

From (a), we also have

- (iii) $J(t) = t \frac{d}{dt} \log H(t)$.

With $h_0 = 1$ and $h_r = 2$ for $r \geq 1$, summing the geometric series gives the closed form

$$H(t) = 1 + 2 \sum_{r \geq 1} t^r = 1 + \frac{2t}{1-t} = \frac{1+t}{1-t}.$$

This is the object of study below. For comparison with the deformed family we record that

$$\frac{d}{dt} \log H(t) = \frac{d}{dt} (\log(1+t) - \log(1-t)) = \frac{1}{1+t} + \frac{1}{1-t} = \frac{2}{1-t^2},$$

so $J(t) = t \cdot \frac{2}{1-t^2} = 2t + 2t^3 + 2t^5 + \dots$, i.e. $j_r = 2$ for odd r and $j_r = 0$ for even r . The deformed analysis prepends a parameter c to this j -sequence and arrives at $H^{(c)}(t) = e^{ct}/(1-t^2)$; the present note keeps J as it is and analyses $H(t) = (1+t)/(1-t)$ directly. We will see that the two differ in exactly one structural respect — the singularity at $t = -1$ — and that this single difference is the whole of the oscillation phenomenon.

We give a self-contained analytic derivation of the large- n behavior of the coefficients h_n of the undeformed generating function $H(t) = (1+t)/(1-t)$. The only input is the closed form on the unit disc; no individual coefficient is computed and no numerical evidence is invoked. We obtain an exact formula

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for h_n , show that the sequence is *eventually exactly constant* rather than merely convergent, and obtain the single limit 2 with no parity splitting. A closing section compares this with the deformed family $H^{(c)}(t) = e^{ct}/(1-t^2)$, whose coefficients split into $\cosh c$ and $\sinh c$, and isolates the simple pole at $t = -1$ as the unique source of that oscillation.

1. SETUP AND NOTATION

We take as our single starting point the identity, valid as an equality of analytic functions on the open unit disc $\{t \in \mathbb{C} : |t| < 1\}$,

$$(1) \quad H(t) = \frac{1+t}{1-t} = \sum_{n=0}^{\infty} h_n t^n.$$

This is what equation (D) produces for the constant sequence $h_0 = 1$, $h_n = 2$ ($n \geq 1$) *before any deformation*: summing the geometric series gives the rational function on the left, and the two sides, being analytic on the unit disc, agree throughout it. The numbers h_n are exactly the Taylor coefficients in (1); their large- n behavior is the object of study. Nothing below uses the value of any particular h_n .

Throughout, “boundary” refers to the circle $|t| = 1$, the boundary of the disc of convergence in (1). The only point of $\overline{\{|t| < 1\}}$ at which H fails to be analytic is $t = +1$; the point $t = -1$, where the denominator $1 - t$ does not vanish, is not a singularity. This asymmetry is the entire content of what follows.

2. AN EXACT FORMULA FOR THE COEFFICIENTS

Lemma 2.1. *For every $n \geq 0$,*

$$h_n = 2 - \delta_{n,0} \quad \text{equivalently} \quad h_0 = 1, \quad h_n = 2 \quad (n \geq 1),$$

where $\delta_{n,0}$ is the Kronecker delta.

Proof. We decompose H over the two boundary points $t = \pm 1$, exactly as the deformed calculation decomposes $1/(1-t^2)$. Since H is a ratio of polynomials of equal degree, the decomposition carries a polynomial part:

$$\frac{1+t}{1-t} = \underbrace{\frac{2}{1-t}}_{\text{pole at } t=+1} + \underbrace{\frac{0}{1+t}}_{\text{pole at } t=-1} - 1,$$

an equality of rational functions on $\mathbb{C} \setminus \{1\}$; one checks it by clearing denominators, $2 - (1-t) = 1+t$. The coefficient attached to the second boundary point $t = -1$ is identically zero. On the disc $|t| < 1$ the surviving pole expands as the absolutely convergent geometric series $\frac{1}{1-t} = \sum_{i \geq 0} t^i$, so that

$$[t^n] \frac{2}{1-t} = 2 \quad (n \geq 0), \quad [t^n](-1) = -\delta_{n,0}, \quad [t^n] \frac{0}{1+t} = 0.$$

Since the Taylor coefficients of the analytic function H are unique, comparison with (1) gives $h_n = 2 - \delta_{n,0}$. $\square$

It is worth isolating what the proof used. The expansion $\frac{1}{1+t} = \sum_{i \geq 0} (-1)^i t^i$ is the only place a factor $(-1)^n$ could enter a coefficient of H ; it is attached to the pole at $t = -1$. Here that pole is absent (its coefficient is 0), so no $(-1)^n$ ever appears. In the deformed companion the same expansion is present with coefficient $\frac{1}{2}e^{-c} \neq 0$, and it is precisely that term which alternates.

3. A TERMINATION ESTIMATE (THE ANALOGUE OF THE EXPONENTIAL TAIL)

In the deformed problem the numerator e^{ct} is an infinite series, and the error in the limiting expression is governed by the tail of that series. Here the numerator $1 + t$ is a polynomial, the convolution terminates, and the corresponding “tail” is not merely small but identically zero past the degree of the numerator. We state this in the form we need.

Lemma 3.1. *Let $P(t) = \sum_{k=0}^d p_k t^k$ be a polynomial and $H(t) = P(t)/(1 - t) = \sum_{n \geq 0} h_n t^n$ on $|t| < 1$. Then for every $n \geq d$,*

$$h_n = P(1) = \sum_{k=0}^d p_k,$$

a constant independent of n . In particular the deviation of h_n from this constant is exactly 0 for all $n \geq d$.

Proof. On $|t| < 1$ write $H(t) = P(t) \sum_{m \geq 0} t^m$, a product of an entire polynomial and an absolutely convergent geometric series; the Cauchy product converges absolutely and its t^n coefficient is the finite convolution

$$h_n = \sum_{k=0}^{\min(n,d)} p_k.$$

For $n \geq d$ the upper limit is d , so $h_n = \sum_{k=0}^d p_k = P(1)$, independent of n . $\square$

Corollary 3.2. *For $H(t) = (1 + t)/(1 - t)$ we have $P(t) = 1 + t$, so $d = 1$ and $P(1) = 2$; hence $h_n = 2$ for all $n \geq 1$. Writing $\varepsilon_n := h_n - 2$ for the deviation from the eventual value, we have*

$$\varepsilon_n = -\delta_{n,0}, \quad \text{so} \quad \varepsilon_n = 0 \text{ for all } n \geq 1.$$

This is the structural counterpart of the deformed tail bound $|\varepsilon_n(c)| \leq |c|^{n+1}/(n+1)!e^{|c|}$. There the error decays super-exponentially but is positive for every n ; here it is not merely small, it is zero. The reason is the same in both problems — the size of the error is controlled by how much of the numerator’s expansion has yet to enter the convolution — but a polynomial

numerator exhausts itself after finitely many steps, whereas an exponential numerator never does.

4. THE ASYMPTOTICS

Theorem 4.1. *For every $n \geq 0$,*

$$h_n = 2 + \varepsilon_n, \quad \varepsilon_n = -\delta_{n,0},$$

so $\varepsilon_n = 0$ for all $n \geq 1$. Consequently the full sequence converges,

$$\lim_{n \rightarrow \infty} h_n = 2,$$

and the even and odd subsequences share this single limit,

$$\lim_{\substack{n \rightarrow \infty \\ n \text{ even}}} h_n = \lim_{\substack{n \rightarrow \infty \\ n \text{ odd}}} h_n = 2.$$

There is no parity splitting: the limit does not depend on the parity of n .

Proof. The exact formula is Lemma 2.1, equivalently Corollary 3.2 with $\varepsilon_n = h_n - 2 = -\delta_{n,0}$. For $n \geq 1$ we have $\varepsilon_n = 0$, so $h_n = 2$ identically; a sequence equal to 2 from $n = 1$ onward converges to 2, and every subsequence — in particular the even- n and odd- n subsequences — has the same limit 2. Since no factor $(-1)^n$ occurs (the pole at $t = -1$ being absent, by Lemma 2.1), there is nothing to distinguish the two parities. $\square$

Corollary 4.2 (contrast with the deformed family). *In the deformed companion, for $H^{(c)}(t) = e^{ct}/(1-t^2)$ one has $h_n^{(c)} = \frac{1}{2}(e^c + (-1)^n e^{-c}) + \varepsilon_n(c)$, with even- n limit $\cosh c$ and odd- n limit $\sinh c$, two values that coincide only when $\sinh c = 0$, i.e. $c = 0$. By contrast the undeformed sequence has a single limit 2 for both parities, for every realisation of the construction. The deformed even and odd limits do not specialise to the undeformed value: as $c \rightarrow 0$ they tend to 1 and 0, the coefficients of $1/(1-t^2)$, not to 2. The undeformed sequence $(1+t)/(1-t)$ is therefore not a member of the one-parameter deformed family $e^{ct}/(1-t^2)$, and the comparison below is between two genuinely different generating functions, not two values of one parameter.*

5. COMPARISON WITH THE DEFORMED FAMILY: LOCATING THE OSCILLATION

Remark 1 (which pole supplies what). Place the two generating functions side by side on the closed unit disc and list their singularities on the boundary circle $|t| = 1$:

$$H^{(c)}(t) = \frac{e^{ct}}{(1-t)(1+t)} \quad (\text{poles at } t = +1 \text{ and } t = -1),$$

$$H(t) = \frac{1+t}{1-t} \quad (\text{pole at } t = +1 \text{ only}).$$

In each case the coefficient asymptotics are governed by these boundary poles. A simple pole at $t = +1$ contributes a constant (parity-independent) mode to h_n ; a simple pole at $t = -1$ contributes the mode $(-1)^n$, because $\frac{1}{1+t} = \sum_n (-1)^n t^n$.

For the deformed function both poles are present and neither is cancelled, since the numerator e^{ct} is entire and zero-free. The pole at $t = +1$ supplies the constant $\frac{1}{2}e^c$ and the pole at $t = -1$ supplies the alternating $\frac{1}{2}(-1)^n e^{-c}$; their sum is $\cosh c$ on even n and $\sinh c$ on odd n . The splitting of the limit into two distinct values — the oscillation — is produced *solely* by the *location* $t = -1$ of the second pole, and the parameter c only fixes the two residue magnitudes $e^{\pm c}/2$.

For the undeformed function the numerator $1 + t$ vanishes at $t = -1$ *exactly where the second pole would sit*, cancelling it. Only the pole at $t = +1$ survives, only the constant mode is present, and the alternating mode has coefficient 0 (Lemma 2.1). Hence a single limit and no oscillation. The contrast between oscillation and its absence is thus, quite literally, the presence or absence of a singularity at the single point $t = -1$.

Remark 2 (a one-line criterion). The mechanism is general. If G is meromorphic on a neighborhood of $\{|t| < 1\}$ with its only boundary singularities simple poles at points $e^{i\theta_1}, \dots, e^{i\theta_k}$, then $[t^n]G$ is asymptotically a sum of terms $\rho_j e^{-in\theta_j}$, one per pole. The sequence converges to a single limit (no oscillation) precisely when the only boundary pole sits at $\theta = 0$, i.e. at $t = +1$; any boundary pole with $\theta \neq 0$ contributes a genuinely oscillating mode of angular frequency θ . The value $\theta = \pi$, i.e. a pole at $t = -1$, gives the period-2 mode $(-1)^n$ and hence the even/odd dichotomy seen in the deformed coefficients. In one sentence: *oscillation in the coefficient sequence is the signature of a dominant boundary singularity off the positive real axis*. The deformation installs such a singularity at $t = -1$ (via the factor $1 - t^2$, which the zero-free numerator e^{ct} cannot repair); the undeformed sequence removes it (via the numerator zero of $1 + t$ at $t = -1$).

Remark 3 (relation to the spectral plots). The oscillation studied empirically in the main article lives one level further on: it is in the minimum-modulus eigenvalues μ_n of the matrices $J_{h,n}^{(c)}$, not directly in the coefficients h_n . The present note does not prove anything about μ_n . What it does supply is the analytic shadow of that phenomenon at the level of the defining sequence: a period-2 $((-1)^n)$ structure is present in the deformed coefficients and absent in the undeformed ones, for exactly the reason given in Remark 1. This is consistent with the observation recorded for the 2-sequence (the deformed $c = 1$ plot, with its even subsequence approaching a constant and its odd subsequence sitting at a separate value), and with the flatness of the undeformed picture. We flag the link as heuristic: whether the boundary-pole dichotomy propagates through the determinant and characteristic-polynomial machinery to control the μ_n is a separate question, and is not settled here.

Remark 4 (independence from numerics). Every assertion follows from (1) together with the elementary Lemmas 2.1 and 3.1. In particular the conclusion “no oscillation” is not an extrapolation from computed coefficients: for each $n \geq 1$ the coefficient h_n is shown to equal 2 exactly, and the deviation ε_n is shown to vanish identically, with no appeal to a table of values.

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